MY_LS_

< LIST DIRECTORY CONTENT />

MY_LS

binary name: my_ls
language: C
compilation: via Makefile, including re, clean and fclean rules

- ✓ The totality of your source files, except all useless files (binary, temp files, obj files,...), must be included in your delivery.
- ✓ All the bonus files (including a potential specific Makefile) should be in a directory named bonus.
- ✓ Error messages have to be written on the error output, and the program should then exit with the 84 error code (0 if there is no error).

Lore
A challenging task for our first mission.
We have been mandated to recode the infamous **ls** binary.
Quite possibly the most used core utils command...

{EPITECH}

The project

You must recode the **ls [OPTION]... [FILE]...** command.

[OPTION]...
Here are the options you must handle: **-alRdt**.
Options, if any, can be given in any order.
Those commands are equivalent:

```
Terminal
$>  ls -alRdt
...
$> ls -ltaRd
...
$> ls -d -t -a -d -l -R
...
```

[FILE]...
Files arguments, if any, must be handled after the options.

```
Terminal
$>  ./my_ls file1
...
$> ./my_ls -l file1 file2
...
$> ./my_ls file1 file2 /dev
...
$> ./my_ls -l -a /usr/sbin
...
```

Examples

```
$>  ./my_ls .
bin include logs Makefile obj src tests todo
```

```
$>  ./my_ls -ld /tmp
drwxrwxrwt 20 root root 20480 Sep 26 13:44 /tmp
```

```
$>  ./my_ls -l /dev | tail -n 10
crw-rw---- 1 root tty 7, 71 Sep 26 09:49 vcsu7
drwxr-xr-x 2 root root 60 Sep 26 09:49 vfio
crw------- 1 root root 10, 127 Sep 26 09:49 vga_arbiter
crw------- 1 root root 10, 137 Sep 26 09:49 vhci
crw-rw---- 1 root kvm 10, 238 Sep 26 09:49 vhost-net
crw-rw---- 1 root kvm 10, 241 Sep 26 09:49 vhost-vsock
crw-rw---- 1 root video 81, 0 Sep 26 09:49 video0
crw-rw---- 1 root video 81, 1 Sep 26 09:49 video1
crw-rw-rw- 1 root root 1, 5 Sep 26 09:49 zero
crw------- 1 root root 10, 249 Sep 26 09:49 zfs
```

```
$>  ./my_ls -lR
.:
total 32
drwxr-xr-x 2 user group 4096 Feb 7 2025 bin
drwxr-xr-x 2 user group 4096 Nov 6 2022 include
drwxr-xr-x 2 user group 4096 Nov 6 2022 logs
-rw-r--r-- 1 user group 1279 Nov 21 2023 Makefile
drwxr-xr-x 2 user group 4096 Sep 26 13:57 obj
drwxr-xr-x 2 user group 4096 Nov 6 2022 src
drwxr-xr-x 3 user group 4096 Nov 20 2022 tests
-rw-r--r-- 1 user group 470 Nov 20 2022 todo
./bin:
total 20
-rw-r--r-- 1 user group 17240 Feb 7 2025 my_ls
./include:
... etc
```

Authorized Functions

The only system calls and libc function allowed are the following ones:

- ✓ opendir, readdir, closedir, stat, lstat, write,
- ✓ getpwuid, getgrgid, time, ctime, readlink,
- ✓ va_start, va_arg, va_end,
- ✓ malloc, free, perror, strerror, exit
- ✓ gnu_dev_major, gnu_dev_minor

Bonus
Unlike the real ls, column formatting is not required and should be treated as a bonus.
The default alphabetical sorting should be treated as a bonus.
Colors should be treated as a bonus.
Bonus ideas: options -r -h -u -S

Some systems show additional permission features:

+ ACL (Access Control List)
. SELinux context
@ Extended file attributes

This is outside the scope of the project and **is not to be handled**.

Terminal

```
$>  ls -l /dev | grep '+'
crw-rw----+ 1 root kvm 10, 232 Oct 29 12:13 kvm
crw-rw----+ 1 root video 234, 0 Oct 29 12:13 media0
crw-rw-r--+ 1 root root 10, 242 Oct 29 12:13 rfkill
crw-rw----+ 1 root video 81, 0 Oct 29 12:13 video0
crw-rw----+ 1 root video 81, 1 Oct 29 12:13 video1
```

v 2.5